

# Bd. 14 - Paarinfima und Paarsuprema

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# Kapitel 1

## Einleitung

Dieser Band spezialisiert die allgemeinen Infimums- und Supremumsbegriffe aus Band 13 auf zweielementige Mengen. Er beweist ihre universellen Eigenschaften und algebraischen Gesetze und behandelt die Inklusionsordnung auf Mengenfamilien als kanonisches Beispiel. Damit entsteht die ordnungstheoretische Grundlage für die späteren Halbverbände und Verbände.

## Kapitel 2

# Paarinfima und Paarsuprema

### 2.1 Existenz und universelle Eigenschaften

Für zwei Elemente werden die allgemeinen Auswertungssätze besonders übersichtlich. In den folgenden Aussagen wird die Existenz des jeweiligen Paarmengensupremums beziehungsweise Paarmengeninfimums ausdrücklich vorausgesetzt.

**Definition 14.2.1.1 (Existenz eines Paarmengensupremums).**

*Mengen:*  $A, x, y, s, u$   
*Relationsoperatoren:*  $\leq$   
*Partielle Ordnung:*  $\triangleright \text{PartOrd}(A, \leq)$   
*Neues Symbol:*  
*Existenzprädikat:*  $\triangleright \text{SupEx}_{A, \leq}(x, y)$

$$\text{SupEx}_{A, \leq}(x, y) := x, y \in A \wedge \exists s \in \text{UB}_{A, \leq}(\{x, y\}) \forall u \in \text{UB}_{A, \leq}(\{x, y\}) (s \leq u)$$

**Definition 14.2.1.2 (Existenz eines Paarmengeninfimums).**

*Mengen:*  $A, x, y, i, d$   
*Relationsoperatoren:*  $\leq$   
*Partielle Ordnung:*  $\triangleright \text{PartOrd}(A, \leq)$   
*Neues Symbol:*  
*Existenzprädikat:*  $\triangleright \text{InfEx}_{A, \leq}(x, y)$

$$\text{InfEx}_{A, \leq}(x, y) := x, y \in A \wedge \exists i \in \text{LB}_{A, \leq}(\{x, y\}) \forall d \in \text{LB}_{A, \leq}(\{x, y\}) (d \leq i)$$

**Theorem 14.2.1.1 (Erster Operand liegt unter dem Paarmengensupremum).**

**Mengen:**  $A, x, y, s, u$   
**Relationsoperatoren:**  $\leq$   
**Partielle Ordnung:**  $\triangleright \text{PartOrd}(A, \leq)$

$$x, y \in A, \exists s \in \text{UB}_{A, \leq}(\{x, y\}) \forall u \in \text{UB}_{A, \leq}(\{x, y\}) (s \leq u) \vdash$$

$$x \leq \sup_{A, \leq}(\{x, y\})$$

|       |   |                                                                                                      |                        |
|-------|---|------------------------------------------------------------------------------------------------------|------------------------|
| 1     | 1 | $x \in A$                                                                                            | $A$                    |
| 2     | 2 | $y \in A$                                                                                            | $A$                    |
| 3     | 3 | $\exists s \in \text{UB}_{A, \leq}(\{x, y\}) \forall u \in \text{UB}_{A, \leq}(\{x, y\}) (s \leq u)$ | $A$                    |
| 1,2   | 4 | $\{x, y\} \subseteq A$                                                                               | Th. 3.13.3.74(1, 2)    |
|       | 5 | $x \in \{x, y\}$                                                                                     | Th. 3.11.3.3           |
| 1,2,3 | 6 | $x \leq \sup_{A, \leq}(\{x, y\})$                                                                    | Th. 13.2.1.13(4, 3, 5) |

**Theorem 14.2.1.2 (Zweiter Operand liegt unter dem Paarmengensupremum).**

**Mengen:**  $A, x, y, s, u$   
**Relationsoperatoren:**  $\leq$   
**Partielle Ordnung:**  $\triangleright \text{PartOrd}(A, \leq)$

$$x, y \in A, \exists s \in \text{UB}_{A, \leq}(\{x, y\}) \forall u \in \text{UB}_{A, \leq}(\{x, y\}) (s \leq u) \vdash$$

$$y \leq \sup_{A, \leq}(\{x, y\})$$

|       |   |                                                                                                      |                        |
|-------|---|------------------------------------------------------------------------------------------------------|------------------------|
| 1     | 1 | $x \in A$                                                                                            | $A$                    |
| 2     | 2 | $y \in A$                                                                                            | $A$                    |
| 3     | 3 | $\exists s \in \text{UB}_{A, \leq}(\{x, y\}) \forall u \in \text{UB}_{A, \leq}(\{x, y\}) (s \leq u)$ | $A$                    |
| 1,2   | 4 | $\{x, y\} \subseteq A$                                                                               | Th. 3.13.3.74(1, 2)    |
|       | 5 | $y \in \{x, y\}$                                                                                     | Th. 3.11.3.4           |
| 1,2,3 | 6 | $y \leq \sup_{A, \leq}(\{x, y\})$                                                                    | Th. 13.2.1.13(4, 3, 5) |

**Theorem 14.2.1.3 (Universalcharakterisierung des Paarmengensupremums).**

**Mengen:**  $A, x, y, u, s, v$   
**Relationsoperatoren:**  $\leq$   
**Partielle Ordnung:**  $\triangleright \text{PartOrd}(A, \leq)$

$$x, y, u \in A,$$

$$\exists s \in \text{UB}_{A, \leq}(\{x, y\}) \forall v \in \text{UB}_{A, \leq}(\{x, y\}) (s \leq v),$$

$$x \leq u, y \leq u \vdash$$

$$\sup_{A, \leq}(\{x, y\}) \leq u$$

|                 |    |                                                                                                                  |                                     |
|-----------------|----|------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 1               | 1  | $x \in A$                                                                                                        | $A$                                 |
| 2               | 2  | $y \in A$                                                                                                        | $A$                                 |
| 3               | 3  | $u \in A$                                                                                                        | $A$                                 |
| 4               | 4  | $\exists s \in \text{UB}_{A,\leq}(\{x, y\}) ($<br>$\quad \forall v \in \text{UB}_{A,\leq}(\{x, y\}) (s \leq v))$ | $A$                                 |
| 5               | 5  | $x \leq u$                                                                                                       | $A$                                 |
| 6               | 6  | $y \leq u$                                                                                                       | $A$                                 |
| 1,2             | 7  | $\{x, y\} \subseteq A$                                                                                           | Th. 3.13.3.74(1, 2)                 |
| 5,6             | 8  | $x \leq u \wedge y \leq u$                                                                                       | $\wedge I(5, 6)$                    |
| 1,2             | 9  | $u \in \text{UB}_{A,\leq}(\{x, y\}) \leftrightarrow (u \in A \wedge x \leq u \wedge y \leq u)$                   | Th. 13.2.1.9(1, 2)                  |
| 1,2,3,5,6       | 10 | $u \in \text{UB}_{A,\leq}(\{x, y\})$                                                                             | Th. 2.2.4.2(9, $\wedge I(3,$<br>8)) |
| 1,2,3,4,5,<br>6 | 11 | $\sup_{A,\leq}(\{x, y\}) \leq u$                                                                                 | Th. 13.2.1.14(7, 4,<br>10)          |

**Theorem 14.2.1.4 (Erster Operand liegt über dem Paarmengeninfimum).**

**Mengen:**  $A, x, y, i, d$   
**Relationsoperatoren:**  $\leq$   
**Partielle Ordnung:**  $\triangleright \text{PartOrd}(A, \leq)$

$$x, y \in A, \exists i \in \text{LB}_{A,\leq}(\{x, y\}) \forall d \in \text{LB}_{A,\leq}(\{x, y\}) (d \leq i) \vdash \\ \inf_{A,\leq}(\{x, y\}) \leq x$$

|       |   |                                                                                                    |                        |
|-------|---|----------------------------------------------------------------------------------------------------|------------------------|
| 1     | 1 | $x \in A$                                                                                          | $A$                    |
| 2     | 2 | $y \in A$                                                                                          | $A$                    |
| 3     | 3 | $\exists i \in \text{LB}_{A,\leq}(\{x, y\}) \forall d \in \text{LB}_{A,\leq}(\{x, y\}) (d \leq i)$ | $A$                    |
| 1,2   | 4 | $\{x, y\} \subseteq A$                                                                             | Th. 3.13.3.74(1, 2)    |
|       | 5 | $x \in \{x, y\}$                                                                                   | Th. 3.11.3.3           |
| 1,2,3 | 6 | $\inf_{A,\leq}(\{x, y\}) \leq x$                                                                   | Th. 13.2.1.19(4, 3, 5) |

**Theorem 14.2.1.5 (Zweiter Operand liegt über dem Paarmengeninfimum).**

**Mengen:**  $A, x, y, i, d$   
**Relationsoperatoren:**  $\leq$   
**Partielle Ordnung:**  $\triangleright \text{PartOrd}(A, \leq)$

$$x, y \in A, \exists i \in \text{LB}_{A,\leq}(\{x, y\}) \forall d \in \text{LB}_{A,\leq}(\{x, y\}) (d \leq i) \vdash \\ \inf_{A,\leq}(\{x, y\}) \leq y$$

|       |   |                                                                                                    |                        |
|-------|---|----------------------------------------------------------------------------------------------------|------------------------|
| 1     | 1 | $x \in A$                                                                                          | $A$                    |
| 2     | 2 | $y \in A$                                                                                          | $A$                    |
| 3     | 3 | $\exists i \in \text{LB}_{A,\leq}(\{x, y\}) \forall d \in \text{LB}_{A,\leq}(\{x, y\}) (d \leq i)$ | $A$                    |
| 1,2   | 4 | $\{x, y\} \subseteq A$                                                                             | Th. 3.13.3.74(1, 2)    |
|       | 5 | $y \in \{x, y\}$                                                                                   | Th. 3.11.3.4           |
| 1,2,3 | 6 | $\inf_{A,\leq}(\{x, y\}) \leq y$                                                                   | Th. 13.2.1.19(4, 3, 5) |

**Theorem 14.2.1.6 (Universalcharakterisierung des Paarmengeninfinimums).**

**Mengen:**  $A, x, y, d, i, e$   
**Relationsoperatoren:**  $\leq$   
**Partielle Ordnung:**  $\triangleright \text{PartOrd}(A, \leq)$

$$\begin{aligned}
& x, y, d \in A, \\
& \exists i \in \text{LB}_{A,\leq}(\{x, y\}) \forall e \in \text{LB}_{A,\leq}(\{x, y\}) (e \leq i), \\
& d \leq x, d \leq y \vdash \\
& d \leq \inf_{A,\leq}(\{x, y\})
\end{aligned}$$

|             |    |                                                                                                |                                   |
|-------------|----|------------------------------------------------------------------------------------------------|-----------------------------------|
| 1           | 1  | $x \in A$                                                                                      | $A$                               |
| 2           | 2  | $y \in A$                                                                                      | $A$                               |
| 3           | 3  | $d \in A$                                                                                      | $A$                               |
| 4           | 4  | $\exists i \in \text{LB}_{A,\leq}(\{x, y\}) ($                                                 | $A$                               |
|             |    | $\forall e \in \text{LB}_{A,\leq}(\{x, y\}) (e \leq i))$                                       |                                   |
| 5           | 5  | $d \leq x$                                                                                     | $A$                               |
| 6           | 6  | $d \leq y$                                                                                     | $A$                               |
| 1,2         | 7  | $\{x, y\} \subseteq A$                                                                         | Th. 3.13.3.74(1, 2)               |
| 5,6         | 8  | $d \leq x \wedge d \leq y$                                                                     | $\wedge I(5, 6)$                  |
| 1,2         | 9  | $d \in \text{LB}_{A,\leq}(\{x, y\}) \leftrightarrow (d \in A \wedge d \leq x \wedge d \leq y)$ | Th. 13.2.1.10(1, 2)               |
| 1,2,3,5,6   | 10 | $d \in \text{LB}_{A,\leq}(\{x, y\})$                                                           | Th. 2.2.4.2(9, $\wedge I(3, 8)$ ) |
| 1,2,3,4,5,6 | 11 | $d \leq \inf_{A,\leq}(\{x, y\})$                                                               | Th. 13.2.1.20(7, 4, 10)           |

## 2.2 Algebraische Gesetze

**Theorem 14.2.2.1 (Idempotenz des Paarmengensupremums).**

**Mengen:**  $A, x, u$   
**Relationsoperatoren:**  $\leq$   
**Partielle Ordnung:**  $\triangleright \text{PartOrd}(A, \leq)$

$$x \in A \vdash \sup_{A,\leq}(\{x, x\}) = x$$

|     |    |                                                                                                 |                                                  |
|-----|----|-------------------------------------------------------------------------------------------------|--------------------------------------------------|
| 1   | 1  | $x \in A$                                                                                       | $A$                                              |
| 1   | 2  | $x \leq x$                                                                                      | Ax. 11.2.1.1(1)                                  |
| 1   | 3  | $\{x, x\} \subseteq A$                                                                          | Th. 3.13.3.74(1, 1)                              |
| 1   | 4  | $x \in A \wedge x \leq x \wedge x \leq x$                                                       | $\wedge I(1, \wedge I^*$<br>$(\text{Wdh}_2(2)))$ |
| 1   | 5  | $x \in \text{UB}_{A, \leq}(\{x, x\}) \leftrightarrow (x \in A \wedge x \leq x \wedge x \leq x)$ | Th. 13.2.1.9(1, 1)                               |
| 1   | 6  | $x \in \text{UB}_{A, \leq}(\{x, x\})$                                                           | Th. 2.2.4.2(5, 4)                                |
| 7   | 7  | $u \in \text{UB}_{A, \leq}(\{x, x\})$                                                           | $A$                                              |
| 1   | 8  | $u \in \text{UB}_{A, \leq}(\{x, x\}) \leftrightarrow (u \in A \wedge x \leq u \wedge x \leq u)$ | Th. 13.2.1.9(1, 1)                               |
| 1,7 | 9  | $u \in A \wedge x \leq u \wedge x \leq u$                                                       | Th. 2.2.4.1(8, 7)                                |
| 1,7 | 10 | $x \leq u$                                                                                      | $\wedge E1(\wedge E2(9))$                        |
| 1   | 11 | $\forall u \in \text{UB}_{A, \leq}(\{x, x\}) (x \leq u)$                                        | $\forall I(\rightarrow I(7, 10))$                |
| 1   | 12 | $x = \sup_{A, \leq}(\{x, x\})$                                                                  | Th. 13.2.1.15(3, 6,<br>11)                       |
| 1   | 13 | $\sup_{A, \leq}(\{x, x\}) = x$                                                                  | Th. 2.9.1.1(12)                                  |

**Theorem 14.2.2.2 (Idempotenz des Paarmengeninfimums).**

*Mengen:*  $A, x, d$   
*Relationsoperatoren:*  $\leq$   
*Partielle Ordnung:*  $\triangleright \text{PartOrd}(A, \leq)$

$$x \in A \vdash \inf_{A, \leq}(\{x, x\}) = x$$

|     |    |                                                                                                 |                                                  |
|-----|----|-------------------------------------------------------------------------------------------------|--------------------------------------------------|
| 1   | 1  | $x \in A$                                                                                       | $A$                                              |
| 1   | 2  | $x \leq x$                                                                                      | Ax. 11.2.1.1(1)                                  |
| 1   | 3  | $\{x, x\} \subseteq A$                                                                          | Th. 3.13.3.74(1, 1)                              |
| 1   | 4  | $x \in A \wedge x \leq x \wedge x \leq x$                                                       | $\wedge I(1, \wedge I^*$<br>$(\text{Wdh}_2(2)))$ |
| 1   | 5  | $x \in \text{LB}_{A, \leq}(\{x, x\}) \leftrightarrow (x \in A \wedge x \leq x \wedge x \leq x)$ | Th. 13.2.1.10(1, 1)                              |
| 1   | 6  | $x \in \text{LB}_{A, \leq}(\{x, x\})$                                                           | Th. 2.2.4.2(5, 4)                                |
| 7   | 7  | $d \in \text{LB}_{A, \leq}(\{x, x\})$                                                           | $A$                                              |
| 1   | 8  | $d \in \text{LB}_{A, \leq}(\{x, x\}) \leftrightarrow (d \in A \wedge d \leq x \wedge d \leq x)$ | Th. 13.2.1.10(1, 1)                              |
| 1,7 | 9  | $d \in A \wedge d \leq x \wedge d \leq x$                                                       | Th. 2.2.4.1(8, 7)                                |
| 1,7 | 10 | $d \leq x$                                                                                      | $\wedge E1(\wedge E2(9))$                        |
| 1   | 11 | $\forall d \in \text{LB}_{A, \leq}(\{x, x\}) (d \leq x)$                                        | $\forall I(\rightarrow I(7, 10))$                |
| 1   | 12 | $x = \inf_{A, \leq}(\{x, x\})$                                                                  | Th. 13.2.1.21(3, 6,<br>11)                       |
| 1   | 13 | $\inf_{A, \leq}(\{x, x\}) = x$                                                                  | Th. 2.9.1.1(12)                                  |

**Theorem 14.2.2.3 (Kommutativität des Paarmengensupremums).**

**Mengen:**  $A, x, y, s, u$   
**Relationsoperatoren:**  $\leq$   
**Partielle Ordnung:**  $\triangleright \text{PartOrd}(A, \leq)$

$$\begin{aligned}
 & x, y \in A, \exists s \in \text{UB}_{A, \leq}(\{x, y\}) \forall u \in \text{UB}_{A, \leq}(\{x, y\}) (s \leq u) \vdash \\
 & \sup_{A, \leq}(\{x, y\}) = \sup_{A, \leq}(\{y, x\})
 \end{aligned}$$

|         |    |                                                                                                      |                                   |
|---------|----|------------------------------------------------------------------------------------------------------|-----------------------------------|
| 1       | 1  | $x \in A$                                                                                            | $A$                               |
| 2       | 2  | $y \in A$                                                                                            | $A$                               |
| 3       | 3  | $\exists s \in \text{UB}_{A, \leq}(\{x, y\}) \forall u \in \text{UB}_{A, \leq}(\{x, y\}) (s \leq u)$ | $A$                               |
|         | 4  | $\{x, y\} = \{y, x\}$                                                                                | Th. 3.11.3.6                      |
| 1,2     | 5  | $\{x, y\} \subseteq A$                                                                               | Th. 3.13.3.74(1, 2)               |
| 1,2,3   | 6  | $\sup_{A, \leq}(\{x, y\}) \in \text{UB}_{A, \leq}(\{x, y\})$                                         | Th. 13.2.1.11(5, 3)               |
| 1,2,3   | 7  | $\sup_{A, \leq}(\{x, y\}) \in \text{UB}_{A, \leq}(\{y, x\})$                                         | $= E(4, 6)$                       |
| 8       | 8  | $u \in \text{UB}_{A, \leq}(\{y, x\})$                                                                | $A$                               |
| 8       | 9  | $u \in \text{UB}_{A, \leq}(\{x, y\})$                                                                | $= E(4, 8)$                       |
| 1,2,3,8 | 10 | $\sup_{A, \leq}(\{x, y\}) \leq u$                                                                    | Th. 13.2.1.14(5, 3, 9)            |
| 1,2,3   | 11 | $\forall u \in \text{UB}_{A, \leq}(\{y, x\}) (\sup_{A, \leq}(\{x, y\}) \leq u)$                      | $\forall I(\rightarrow I(8, 10))$ |
| 1,2     | 12 | $\{y, x\} \subseteq A$                                                                               | Th. 3.13.3.74(2, 1)               |
| 1,2,3   | 13 | $\sup_{A, \leq}(\{x, y\}) = \sup_{A, \leq}(\{y, x\})$                                                | Th. 13.2.1.15(12, 7, 11)          |

**Theorem 14.2.2.4 (Kommutativität des Paarmengeninfimums).**

**Mengen:**  $A, x, y, i, d$   
**Relationsoperatoren:**  $\leq$   
**Partielle Ordnung:**  $\triangleright \text{PartOrd}(A, \leq)$

$$\begin{aligned}
 & x, y \in A, \exists i \in \text{LB}_{A, \leq}(\{x, y\}) \forall d \in \text{LB}_{A, \leq}(\{x, y\}) (d \leq i) \vdash \\
 & \inf_{A, \leq}(\{x, y\}) = \inf_{A, \leq}(\{y, x\})
 \end{aligned}$$

|       |   |                                                                                                      |                     |
|-------|---|------------------------------------------------------------------------------------------------------|---------------------|
| 1     | 1 | $x \in A$                                                                                            | $A$                 |
| 2     | 2 | $y \in A$                                                                                            | $A$                 |
| 3     | 3 | $\exists i \in \text{LB}_{A, \leq}(\{x, y\}) \forall d \in \text{LB}_{A, \leq}(\{x, y\}) (d \leq i)$ | $A$                 |
|       | 4 | $\{x, y\} = \{y, x\}$                                                                                | Th. 3.11.3.6        |
| 1,2   | 5 | $\{x, y\} \subseteq A$                                                                               | Th. 3.13.3.74(1, 2) |
| 1,2,3 | 6 | $\inf_{A, \leq}(\{x, y\}) \in \text{LB}_{A, \leq}(\{x, y\})$                                         | Th. 13.2.1.17(5, 3) |
| 1,2,3 | 7 | $\inf_{A, \leq}(\{x, y\}) \in \text{LB}_{A, \leq}(\{y, x\})$                                         | $= E(4, 6)$         |
| 8     | 8 | $d \in \text{LB}_{A, \leq}(\{y, x\})$                                                                | $A$                 |
| 8     | 9 | $d \in \text{LB}_{A, \leq}(\{x, y\})$                                                                | $= E(4, 8)$         |

|         |    |                                                                               |                                   |
|---------|----|-------------------------------------------------------------------------------|-----------------------------------|
| 1,2,3,8 | 10 | $d \leq \inf_{A,\leq}(\{x, y\})$                                              | Th. 13.2.1.20(5, 3, 9)            |
| 1,2,3   | 11 | $\forall d \in \text{LB}_{A,\leq}(\{y, x\}) (d \leq \inf_{A,\leq}(\{x, y\}))$ | $\forall I(\rightarrow I(8, 10))$ |
| 1,2     | 12 | $\{y, x\} \subseteq A$                                                        | Th. 3.13.3.74(2, 1)               |
| 1,2,3   | 13 | $\inf_{A,\leq}(\{x, y\}) = \inf_{A,\leq}(\{y, x\})$                           | Th. 13.2.1.21(12, 7, 11)          |

**Theorem 14.2.2.5 (Assoziativität des Paarmengensupremums).**

**Mengen:**  $A, x, y, z$   
**Relationsoperatoren:**  $\leq$   
**Partielle Ordnung:**  $\triangleright \text{PartOrd}(A, \leq)$

$$\begin{aligned} & \text{SupEx}_{A,\leq}(x, y), \text{SupEx}_{A,\leq}(y, z) \\ & , \text{SupEx}_{A,\leq}(\sup_{A,\leq}(\{x, y\}), z), \text{SupEx}_{A,\leq}(x, \sup_{A,\leq}(\{y, z\})) \vdash \\ & \sup_{A,\leq}(\sup_{A,\leq}(\{x, y\}), z) = \sup_{A,\leq}(x, \sup_{A,\leq}(\{y, z\})) \end{aligned}$$

*Beweis.*

Träger und Existenz der Suprema:

|   |    |                                                                                                                                                                                                                |                            |
|---|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 1 | 1  | $\text{SupEx}_{A,\leq}(x, y)$                                                                                                                                                                                  | $A$                        |
| 2 | 2  | $\text{SupEx}_{A,\leq}(y, z)$                                                                                                                                                                                  | $A$                        |
| 3 | 3  | $\text{SupEx}_{A,\leq}(\sup_{A,\leq}(\{x, y\}), z)$                                                                                                                                                            | $A$                        |
| 4 | 4  | $\text{SupEx}_{A,\leq}(x, \sup_{A,\leq}(\{y, z\}))$                                                                                                                                                            | $A$                        |
| 1 | 5  | $(x \in A \wedge y \in A) \wedge$<br>$\exists s \in \text{UB}_{A,\leq}(\{x, y\})$<br>$\forall u \in \text{UB}_{A,\leq}(\{x, y\}) (s \leq u)$                                                                   | Def. 14.2.1.1(1)           |
| 1 | 6  | $x \in A$                                                                                                                                                                                                      | $\wedge E1(\wedge E1(5))$  |
| 1 | 7  | $y \in A$                                                                                                                                                                                                      | $\wedge E2(\wedge E1(5))$  |
| 1 | 8  | $\exists s \in \text{UB}_{A,\leq}(\{x, y\}) \forall u \in \text{UB}_{A,\leq}(\{x, y\}) (s \leq u)$                                                                                                             | $\wedge E2(5)$             |
| 2 | 9  | $(y \in A \wedge z \in A) \wedge$<br>$\exists s \in \text{UB}_{A,\leq}(\{y, z\})$<br>$\forall u \in \text{UB}_{A,\leq}(\{y, z\}) (s \leq u)$                                                                   | Def. 14.2.1.1(2)           |
| 2 | 10 | $z \in A$                                                                                                                                                                                                      | $\wedge E2(\wedge E1(9))$  |
| 2 | 11 | $\exists s \in \text{UB}_{A,\leq}(\{y, z\}) \forall u \in \text{UB}_{A,\leq}(\{y, z\}) (s \leq u)$                                                                                                             | $\wedge E2(9)$             |
| 3 | 12 | $(\sup_{A,\leq}(\{x, y\}) \in A \wedge z \in A) \wedge$<br>$\exists s \in \text{UB}_{A,\leq}(\{\sup_{A,\leq}(\{x, y\}), z\})$<br>$\forall u \in \text{UB}_{A,\leq}(\{\sup_{A,\leq}(\{x, y\}), z\}) (s \leq u)$ | Def. 14.2.1.1(3)           |
| 3 | 13 | $\sup_{A,\leq}(\{x, y\}) \in A$                                                                                                                                                                                | $\wedge E1(\wedge E1(12))$ |

|   |    |                                                                                                                                                                                                                     |                            |
|---|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 3 | 14 | $\exists s \in \text{UB}_{A, \leq}(\{\sup_{A, \leq}(\{x, y\}), z\})$                                                                                                                                                | $\wedge E2(12)$            |
|   |    | $\forall u \in \text{UB}_{A, \leq}(\{\sup_{A, \leq}(\{x, y\}), z\}) (s \leq u)$                                                                                                                                     |                            |
| 4 | 15 | $(x \in A \wedge \sup_{A, \leq}(\{y, z\}) \in A) \wedge$<br>$\exists s \in \text{UB}_{A, \leq}(\{x, \sup_{A, \leq}(\{y, z\})\})$<br>$\forall u \in \text{UB}_{A, \leq}(\{x, \sup_{A, \leq}(\{y, z\})\}) (s \leq u)$ | Def. 14.2.1.1(4)           |
| 4 | 16 | $\sup_{A, \leq}(\{y, z\}) \in A$                                                                                                                                                                                    | $\wedge E2(\wedge E1(15))$ |
| 4 | 17 | $\exists s \in \text{UB}_{A, \leq}(\{x, \sup_{A, \leq}(\{y, z\})\})$<br>$\forall u \in \text{UB}_{A, \leq}(\{x, \sup_{A, \leq}(\{y, z\})\}) (s \leq u)$                                                             | $\wedge E2(15)$            |
| 3 | 18 | $\{\sup_{A, \leq}(\{x, y\}), z\} \subseteq A$                                                                                                                                                                       | Th. 3.13.3.74(13, 10)      |
| 3 | 19 | $\sup_{A, \leq}(\{\sup_{A, \leq}(\{x, y\}), z\}) \in A$                                                                                                                                                             | Th. 13.2.1.12(18, 14)      |
| 4 | 20 | $\{x, \sup_{A, \leq}(\{y, z\})\} \subseteq A$                                                                                                                                                                       | Th. 3.13.3.74(6, 16)       |
| 4 | 21 | $\sup_{A, \leq}(\{x, \sup_{A, \leq}(\{y, z\})\}) \in A$                                                                                                                                                             | Th. 13.2.1.12(20, 17)      |

Erste Ungleichung:

|         |    |                                                                                                             |                                          |
|---------|----|-------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 4       | 22 | $x \leq \sup_{A, \leq}(\{x, \sup_{A, \leq}(\{y, z\})\})$                                                    | Th. 14.2.1.1(6, 16, 17)                  |
| 4       | 23 | $\sup_{A, \leq}(\{y, z\}) \leq \sup_{A, \leq}(\{x, \sup_{A, \leq}(\{y, z\})\})$                             | Th. 14.2.1.2(6, 16, 17)                  |
| 2       | 24 | $y \leq \sup_{A, \leq}(\{y, z\})$                                                                           | Th. 14.2.1.1(7, 10, 11)                  |
| 2       | 25 | $z \leq \sup_{A, \leq}(\{y, z\})$                                                                           | Th. 14.2.1.2(7, 10, 11)                  |
| 2,4     | 26 | $y \leq \sup_{A, \leq}(\{x, \sup_{A, \leq}(\{y, z\})\})$                                                    | Ax. 11.2.1.3(7, 16, 21, 24, 23)          |
| 2,4     | 27 | $z \leq \sup_{A, \leq}(\{x, \sup_{A, \leq}(\{y, z\})\})$                                                    | Ax. 11.2.1.3(10, 16, 21, 25, 23)         |
| 1,2,4   | 28 | $\sup_{A, \leq}(\{x, y\}) \leq \sup_{A, \leq}(\{x, \sup_{A, \leq}(\{y, z\})\})$                             | Th. 14.2.1.3<br>(6, 7, 21, 8, 22, 26)    |
| 1,2,3,4 | 29 | $\sup_{A, \leq}(\{\sup_{A, \leq}(\{x, y\}), z\}) \leq$<br>$\sup_{A, \leq}(\{x, \sup_{A, \leq}(\{y, z\})\})$ | Th. 14.2.1.3<br>(13, 10, 21, 14, 28, 27) |

Zweite Ungleichung:

|   |    |                                                                                 |                          |
|---|----|---------------------------------------------------------------------------------|--------------------------|
| 3 | 30 | $\sup_{A, \leq}(\{x, y\}) \leq \sup_{A, \leq}(\{\sup_{A, \leq}(\{x, y\}), z\})$ | Th. 14.2.1.1(13, 10, 14) |
| 3 | 31 | $z \leq \sup_{A, \leq}(\{\sup_{A, \leq}(\{x, y\}), z\})$                        | Th. 14.2.1.2(13, 10, 14) |
| 1 | 32 | $x \leq \sup_{A, \leq}(\{x, y\})$                                               | Th. 14.2.1.1(6, 7, 8)    |
| 1 | 33 | $y \leq \sup_{A, \leq}(\{x, y\})$                                               | Th. 14.2.1.2(6, 7, 8)    |

|         |    |                                                                                                   |                                         |
|---------|----|---------------------------------------------------------------------------------------------------|-----------------------------------------|
| 1,3     | 34 | $x \leq \sup_{A,\leq}(\{\sup_{A,\leq}(\{x,y\}), z\})$                                             | Ax. 11.2.1.3(6, 13, 19, 32, 30)         |
| 1,3     | 35 | $y \leq \sup_{A,\leq}(\{\sup_{A,\leq}(\{x,y\}), z\})$                                             | Ax. 11.2.1.3(7, 13, 19, 33, 30)         |
| 1,2,3   | 36 | $\sup_{A,\leq}(\{y, z\}) \leq \sup_{A,\leq}(\{\sup_{A,\leq}(\{x,y\}), z\})$                       | Th. 14.2.1.3<br>(7, 10, 19, 11, 35, 31) |
| 1,2,3,4 | 37 | $\sup_{A,\leq}(\{x, \sup_{A,\leq}(\{y, z\})\}) \leq \sup_{A,\leq}(\{\sup_{A,\leq}(\{x,y\}), z\})$ | Th. 14.2.1.3<br>(6, 16, 19, 17, 34, 36) |

Gleichheit durch Antisymmetrie:

|         |    |                                                                                               |                      |
|---------|----|-----------------------------------------------------------------------------------------------|----------------------|
| 1,2,3,4 | 38 | $\sup_{A,\leq}(\{\sup_{A,\leq}(\{x,y\}), z\}) = \sup_{A,\leq}(\{x, \sup_{A,\leq}(\{y,z\})\})$ | Ax. 12.2.1.1(29, 37) |
|---------|----|-----------------------------------------------------------------------------------------------|----------------------|

□

**Theorem 14.2.2.6 (Assoziativität des Paarmengeninfimums).**

**Mengen:**  $A, x, y, z$   
**Relationsoperatoren:**  $\leq$   
**Partielle Ordnung:**  $\triangleright \text{PartOrd}(A, \leq)$

$$\begin{aligned} & \text{InfEx}_{A,\leq}(x, y), \text{InfEx}_{A,\leq}(y, z) \\ & , \text{InfEx}_{A,\leq}(\inf_{A,\leq}(\{x, y\}), z), \text{InfEx}_{A,\leq}(x, \inf_{A,\leq}(\{y, z\})) \vdash \\ & \inf_{A,\leq}(\{\inf_{A,\leq}(\{x, y\}), z\}) = \inf_{A,\leq}(\{x, \inf_{A,\leq}(\{y, z\})\}) \end{aligned}$$

*Beweis.*

Träger und Existenz der Infima:

|   |    |                                                                                                                                    |                           |
|---|----|------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| 1 | 1  | $\text{InfEx}_{A,\leq}(x, y)$                                                                                                      | $A$                       |
| 2 | 2  | $\text{InfEx}_{A,\leq}(y, z)$                                                                                                      | $A$                       |
| 3 | 3  | $\text{InfEx}_{A,\leq}(\inf_{A,\leq}(\{x, y\}), z)$                                                                                | $A$                       |
| 4 | 4  | $\text{InfEx}_{A,\leq}(x, \inf_{A,\leq}(\{y, z\}))$                                                                                | $A$                       |
| 1 | 5  | $(x \in A \wedge y \in A) \wedge \exists i \in \text{LB}_{A,\leq}(\{x, y\}) \forall d \in \text{LB}_{A,\leq}(\{x, y\}) (d \leq i)$ | Def. 14.2.1.2(1)          |
| 1 | 6  | $x \in A$                                                                                                                          | $\wedge E1(\wedge E1(5))$ |
| 1 | 7  | $y \in A$                                                                                                                          | $\wedge E2(\wedge E1(5))$ |
| 1 | 8  | $\exists i \in \text{LB}_{A,\leq}(\{x, y\}) \forall d \in \text{LB}_{A,\leq}(\{x, y\}) (d \leq i)$                                 | $\wedge E2(5)$            |
| 2 | 9  | $(y \in A \wedge z \in A) \wedge \exists i \in \text{LB}_{A,\leq}(\{y, z\}) \forall d \in \text{LB}_{A,\leq}(\{y, z\}) (d \leq i)$ | Def. 14.2.1.2(2)          |
| 2 | 10 | $z \in A$                                                                                                                          | $\wedge E2(\wedge E1(9))$ |

|   |    |                                                                                                                                                                                                                |                            |
|---|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 2 | 11 | $\exists i \in \text{LB}_{A,\leq}(\{y, z\}) \forall d \in \text{LB}_{A,\leq}(\{y, z\}) (d \leq i)$                                                                                                             | $\wedge E2(9)$             |
| 3 | 12 | $(\inf_{A,\leq}(\{x, y\}) \in A \wedge z \in A) \wedge$<br>$\exists i \in \text{LB}_{A,\leq}(\{\inf_{A,\leq}(\{x, y\}), z\})$<br>$\forall d \in \text{LB}_{A,\leq}(\{\inf_{A,\leq}(\{x, y\}), z\}) (d \leq i)$ | Def. 14.2.1.2(3)           |
| 3 | 13 | $\inf_{A,\leq}(\{x, y\}) \in A$                                                                                                                                                                                | $\wedge E1(\wedge E1(12))$ |
| 3 | 14 | $\exists i \in \text{LB}_{A,\leq}(\{\inf_{A,\leq}(\{x, y\}), z\})$<br>$\forall d \in \text{LB}_{A,\leq}(\{\inf_{A,\leq}(\{x, y\}), z\}) (d \leq i)$                                                            | $\wedge E2(12)$            |
| 4 | 15 | $(x \in A \wedge \inf_{A,\leq}(\{y, z\}) \in A) \wedge$<br>$\exists i \in \text{LB}_{A,\leq}(\{x, \inf_{A,\leq}(\{y, z\})\})$<br>$\forall d \in \text{LB}_{A,\leq}(\{x, \inf_{A,\leq}(\{y, z\})\}) (d \leq i)$ | Def. 14.2.1.2(4)           |
| 4 | 16 | $\inf_{A,\leq}(\{y, z\}) \in A$                                                                                                                                                                                | $\wedge E2(\wedge E1(15))$ |
| 4 | 17 | $\exists i \in \text{LB}_{A,\leq}(\{x, \inf_{A,\leq}(\{y, z\})\})$<br>$\forall d \in \text{LB}_{A,\leq}(\{x, \inf_{A,\leq}(\{y, z\})\}) (d \leq i)$                                                            | $\wedge E2(15)$            |
| 3 | 18 | $\{\inf_{A,\leq}(\{x, y\}), z\} \subseteq A$                                                                                                                                                                   | Th. 3.13.3.74(13, 10)      |
| 3 | 19 | $\inf_{A,\leq}(\{\inf_{A,\leq}(\{x, y\}), z\}) \in A$                                                                                                                                                          | Th. 13.2.1.18(18, 14)      |
| 4 | 20 | $\{x, \inf_{A,\leq}(\{y, z\})\} \subseteq A$                                                                                                                                                                   | Th. 3.13.3.74(6, 16)       |
| 4 | 21 | $\inf_{A,\leq}(\{x, \inf_{A,\leq}(\{y, z\})\}) \in A$                                                                                                                                                          | Th. 13.2.1.18(20, 17)      |

Erste Ungleichung:

|         |    |                                                                                                         |                                         |
|---------|----|---------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 3       | 22 | $\inf_{A,\leq}(\{\inf_{A,\leq}(\{x, y\}), z\}) \leq \inf_{A,\leq}(\{x, y\})$                            | Th. 14.2.1.4(13, 10, 14)                |
| 3       | 23 | $\inf_{A,\leq}(\{\inf_{A,\leq}(\{x, y\}), z\}) \leq z$                                                  | Th. 14.2.1.5(13, 10, 14)                |
| 1       | 24 | $\inf_{A,\leq}(\{x, y\}) \leq x$                                                                        | Th. 14.2.1.4(6, 7, 8)                   |
| 1       | 25 | $\inf_{A,\leq}(\{x, y\}) \leq y$                                                                        | Th. 14.2.1.5(6, 7, 8)                   |
| 1,3     | 26 | $\inf_{A,\leq}(\{\inf_{A,\leq}(\{x, y\}), z\}) \leq x$                                                  | Ax. 11.2.1.3(19, 13, 6, 22, 24)         |
| 1,3     | 27 | $\inf_{A,\leq}(\{\inf_{A,\leq}(\{x, y\}), z\}) \leq y$                                                  | Ax. 11.2.1.3(19, 13, 7, 22, 25)         |
| 1,2,3   | 28 | $\inf_{A,\leq}(\{\inf_{A,\leq}(\{x, y\}), z\}) \leq \inf_{A,\leq}(\{y, z\})$                            | Th. 14.2.1.6<br>(7, 10, 19, 11, 27, 23) |
| 1,2,3,4 | 29 | $\inf_{A,\leq}(\{\inf_{A,\leq}(\{x, y\}), z\}) \leq$<br>$\inf_{A,\leq}(\{x, \inf_{A,\leq}(\{y, z\})\})$ | Th. 14.2.1.6<br>(6, 16, 19, 17, 26, 28) |

Zweite Ungleichung:

|   |    |                                                        |                         |
|---|----|--------------------------------------------------------|-------------------------|
| 4 | 30 | $\inf_{A,\leq}(\{x, \inf_{A,\leq}(\{y, z\})\}) \leq x$ | Th. 14.2.1.4(6, 16, 17) |
|---|----|--------------------------------------------------------|-------------------------|

|         |    |                                                                                                    |                                          |
|---------|----|----------------------------------------------------------------------------------------------------|------------------------------------------|
| 4       | 31 | $\inf_{A,\leq}(\{x, \inf_{A,\leq}(\{y, z\})\}) \leq \inf_{A,\leq}(\{y, z\})$                       | Th. 14.2.1.5(6, 16, 17)                  |
| 2       | 32 | $\inf_{A,\leq}(\{y, z\}) \leq y$                                                                   | Th. 14.2.1.4(7, 10, 11)                  |
| 2       | 33 | $\inf_{A,\leq}(\{y, z\}) \leq z$                                                                   | Th. 14.2.1.5(7, 10, 11)                  |
| 2,4     | 34 | $\inf_{A,\leq}(\{x, \inf_{A,\leq}(\{y, z\})\}) \leq y$                                             | Ax. 11.2.1.3(21, 16, 7, 31, 32)          |
| 2,4     | 35 | $\inf_{A,\leq}(\{x, \inf_{A,\leq}(\{y, z\})\}) \leq z$                                             | Ax. 11.2.1.3(21, 16, 10, 31, 33)         |
| 1,2,4   | 36 | $\inf_{A,\leq}(\{x, \inf_{A,\leq}(\{y, z\})\}) \leq \inf_{A,\leq}(\{x, y\})$                       | Th. 14.2.1.6<br>(6, 7, 21, 8, 30, 34)    |
| 1,2,3,4 | 37 | $\inf_{A,\leq}(\{x, \inf_{A,\leq}(\{y, z\})\}) \leq \inf_{A,\leq}(\{\inf_{A,\leq}(\{x, y\}), z\})$ | Th. 14.2.1.6<br>(13, 10, 21, 14, 36, 35) |

Gleichheit durch Antisymmetrie:

|         |    |                                                                                                 |                      |
|---------|----|-------------------------------------------------------------------------------------------------|----------------------|
| 1,2,3,4 | 38 | $\inf_{A,\leq}(\{\inf_{A,\leq}(\{x, y\}), z\}) = \inf_{A,\leq}(\{x, \inf_{A,\leq}(\{y, z\})\})$ | Ax. 12.2.1.1(29, 37) |
|---------|----|-------------------------------------------------------------------------------------------------|----------------------|

□

## 2.3 Inklusionsordnung auf Mengenfamilien

**Theorem 14.2.3.1 (Inklusionsordnung auf einer Mengenfamilie).**

**Mengen:**  $U, M, X, Y, Z$   
**Relationsooperatoren:**  $\subseteq$

$$M \subseteq \mathcal{P}(U) \vdash \text{PartOrd}(M, \subseteq)$$

*Beweis.*

Reflexivität:

|   |   |                              |             |
|---|---|------------------------------|-------------|
| 1 | 1 | $M \subseteq \mathcal{P}(U)$ | A           |
| 2 | 2 | $X \in M$                    | A           |
| 3 | 3 | $Y \in M$                    | A           |
| 4 | 4 | $Z \in M$                    | A           |
| 2 | 5 | $X \subseteq X$              | Th. 3.5.3.1 |

Transitivität:

|   |   |                 |   |
|---|---|-----------------|---|
| 6 | 6 | $X \subseteq Y$ | A |
| 7 | 7 | $Y \subseteq Z$ | A |

2,3,4,6,7    8  $X \subseteq Z$

Th. 3.5.3.6(6, 7)

Antisymmetrie:

9    9  $Y \subseteq X$   
2,3,6,9    10  $X = Y$

$A$

Th. 3.5.3.5(6, 9)

Partielle Ordnung:

1    11  $\text{PartOrd}(M, \subseteq)$

Def. 12.2.1.1(5, 8,  
10)

□

Die Reflexivität, Transitivität und Antisymmetrie dieser Ordnung sind genau die entsprechenden Grundgesetze der Teilmengenrelation.

**Theorem 14.2.3.2 (Paarmengeninfimum in einer Mengenfamilie).**

**Mengen:**  $U, M, X, Y, Z$   
**Relationsoperatoren:**  $\subseteq$   
**Partielle Ordnung:**  $\triangleright \text{PartOrd}(M, \subseteq)$

$$M \subseteq \mathcal{P}(U), X, Y \in M, X \cap Y \in M \vdash \\ \inf_{M, \subseteq}(\{X, Y\}) = X \cap Y$$

*Beweis.*

Untere Schranke:

|       |    |                                                                                                                                            |                               |
|-------|----|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 1     | 1  | $M \subseteq \mathcal{P}(U)$                                                                                                               | $A$                           |
|       | 2  | $X \in M$                                                                                                                                  | $A$                           |
|       | 3  | $Y \in M$                                                                                                                                  | $A$                           |
|       | 4  | $X \cap Y \in M$                                                                                                                           | $A$                           |
| 2,3   | 5  | $\{X, Y\} \subseteq M$                                                                                                                     | Th. 3.13.3.74(2, 3)           |
|       | 6  | $X \cap Y \subseteq X$                                                                                                                     | Th. 3.10.2.10                 |
|       | 7  | $X \cap Y \subseteq Y$                                                                                                                     | Th. 3.10.2.11                 |
|       | 8  | $X \cap Y \in M \wedge X \cap Y \subseteq X \wedge X \cap Y \subseteq Y$                                                                   | $\wedge I(4, \wedge I(6, 7))$ |
| 2,3   | 9  | $X \cap Y \in \text{LB}_{M, \subseteq}(\{X, Y\}) \leftrightarrow (X \cap Y \in M \wedge X \cap Y \subseteq X \wedge X \cap Y \subseteq Y)$ | Th. 13.2.1.10(2, 3)           |
| 2,3,4 | 10 | $X \cap Y \in \text{LB}_{M, \subseteq}(\{X, Y\})$                                                                                          | Th. 2.2.4.2(9, 8)             |

Größte untere Schranke:

|        |    |                                                                                                               |                                    |
|--------|----|---------------------------------------------------------------------------------------------------------------|------------------------------------|
| 11     | 11 | $Z \in \text{LB}_{M,\subseteq}(\{X, Y\})$                                                                     | $A$                                |
| 2,3    | 12 | $Z \in \text{LB}_{M,\subseteq}(\{X, Y\}) \leftrightarrow (Z \in M \wedge Z \subseteq X \wedge Z \subseteq Y)$ | Th. 13.2.1.10(2, 3)                |
| 2,3,11 | 13 | $Z \in M \wedge Z \subseteq X \wedge Z \subseteq Y$                                                           | Th. 2.2.4.1(12, 11)                |
| 2,3,11 | 14 | $Z \subseteq X$                                                                                               | $\wedge E1(\wedge E2(13))$         |
| 2,3,11 | 15 | $Z \subseteq Y$                                                                                               | $\wedge E2(\wedge E2(13))$         |
| 2,3,11 | 16 | $Z \subseteq X \cap Y$                                                                                        | Th. 3.10.2.14(14, 15)              |
| 2,3    | 17 | $\forall Z \in \text{LB}_{M,\subseteq}(\{X, Y\}) (Z \subseteq X \cap Y)$                                      | $\forall I(\rightarrow I(11, 16))$ |

Identifikation des Infimums:

|       |    |                                           |                          |
|-------|----|-------------------------------------------|--------------------------|
| 2,3,4 | 18 | $X \cap Y = \inf_{M,\subseteq}(\{X, Y\})$ | Th. 13.2.1.21(5, 10, 17) |
| 2,3,4 | 19 | $\inf_{M,\subseteq}(\{X, Y\}) = X \cap Y$ | Th. 2.9.1.1(18)          |

□

**Theorem 14.2.3.3 (Paarmengensupremum in einer Mengenfamilie).**

**Mengen:**  $U, M, X, Y, Z$   
**Relationsoperatoren:**  $\subseteq$   
**Partielle Ordnung:**  $\triangleright \text{PartOrd}(M, \subseteq)$

$$M \subseteq \mathcal{P}(U), X, Y \in M, X \cup Y \in M \vdash \\ \sup_{M,\subseteq}(\{X, Y\}) = X \cup Y$$

*Beweis.*

Obere Schranke:

|       |    |                                                                                                                                           |                               |
|-------|----|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 1     | 1  | $M \subseteq \mathcal{P}(U)$                                                                                                              | $A$                           |
| 2     | 2  | $X \in M$                                                                                                                                 | $A$                           |
| 3     | 3  | $Y \in M$                                                                                                                                 | $A$                           |
| 4     | 4  | $X \cup Y \in M$                                                                                                                          | $A$                           |
| 2,3   | 5  | $\{X, Y\} \subseteq M$                                                                                                                    | Th. 3.13.3.74(2, 3)           |
|       | 6  | $X \subseteq X \cup Y$                                                                                                                    | Th. 3.13.3.53                 |
|       | 7  | $Y \subseteq X \cup Y$                                                                                                                    | Th. 3.13.3.54                 |
| 4     | 8  | $X \cup Y \in M \wedge X \subseteq X \cup Y \wedge Y \subseteq X \cup Y$                                                                  | $\wedge I(4, \wedge I(6, 7))$ |
| 2,3   | 9  | $X \cup Y \in \text{UB}_{M,\subseteq}(\{X, Y\}) \leftrightarrow (X \cup Y \in M \wedge X \subseteq X \cup Y \wedge Y \subseteq X \cup Y)$ | Th. 13.2.1.9(2, 3)            |
| 2,3,4 | 10 | $X \cup Y \in \text{UB}_{M,\subseteq}(\{X, Y\})$                                                                                          | Th. 2.2.4.2(9, 8)             |

Kleinste obere Schranke:

|        |    |                                                                                                               |                                    |
|--------|----|---------------------------------------------------------------------------------------------------------------|------------------------------------|
| 11     | 11 | $Z \in \text{UB}_{M,\subseteq}(\{X, Y\})$                                                                     | $A$                                |
| 2,3    | 12 | $Z \in \text{UB}_{M,\subseteq}(\{X, Y\}) \leftrightarrow (Z \in M \wedge X \subseteq Z \wedge Y \subseteq Z)$ | Th. 13.2.1.9(2, 3)                 |
| 2,3,11 | 13 | $Z \in M \wedge X \subseteq Z \wedge Y \subseteq Z$                                                           | Th. 2.2.4.1(12, 11)                |
| 2,3,11 | 14 | $X \subseteq Z$                                                                                               | $\wedge E1(\wedge E2(13))$         |
| 2,3,11 | 15 | $Y \subseteq Z$                                                                                               | $\wedge E2(\wedge E2(13))$         |
| 2,3,11 | 16 | $X \cup Y \subseteq Z$                                                                                        | Th. 3.13.3.55(14, 15)              |
| 2,3    | 17 | $\forall Z \in \text{UB}_{M,\subseteq}(\{X, Y\}) (X \cup Y \subseteq Z)$                                      | $\forall I(\rightarrow I(11, 16))$ |

Identifikation des Supremums:

|       |    |                                           |                          |
|-------|----|-------------------------------------------|--------------------------|
| 2,3,4 | 18 | $X \cup Y = \sup_{M,\subseteq}(\{X, Y\})$ | Th. 13.2.1.15(5, 10, 17) |
| 2,3,4 | 19 | $\sup_{M,\subseteq}(\{X, Y\}) = X \cup Y$ | Th. 2.9.1.1(18)          |

□

**Theorem 14.2.3.4 (Paarmengeninfimum im Potenzmengenträger).**

*Mengen:*  $U, X, Y$   
*Relationsoperatoren:*  $\subseteq$

$$X, Y \subseteq U \vdash \inf_{\mathcal{P}(U), \subseteq} (\{X, Y\}) = X \cap Y$$

|     |   |                                                          |                             |
|-----|---|----------------------------------------------------------|-----------------------------|
| 1   | 1 | $X \subseteq U$                                          | $A$                         |
| 2   | 2 | $Y \subseteq U$                                          | $A$                         |
|     | 3 | $\mathcal{P}(U) \subseteq \mathcal{P}(U)$                | Th. 3.5.3.1                 |
| 1   | 4 | $X \in \mathcal{P}(U)$                                   | Th. 3.16.2.1(1)             |
| 2   | 5 | $Y \in \mathcal{P}(U)$                                   | Th. 3.16.2.1(2)             |
| 1   | 6 | $X \cap Y \subseteq U$                                   | Th. 3.10.2.12(1)            |
| 1   | 7 | $X \cap Y \in \mathcal{P}(U)$                            | Th. 3.16.2.1(6)             |
|     | 8 | $\text{PartOrd}(\mathcal{P}(U), \subseteq)$              | Th. 14.2.3.1(3)             |
| 1,2 | 9 | $\inf_{\mathcal{P}(U), \subseteq} (\{X, Y\}) = X \cap Y$ | Th. 14.2.3.2(8, 3, 4, 5, 7) |

**Theorem 14.2.3.5 (Paarmengensupremum im Potenzmengenträger).**

*Mengen:*  $U, X, Y$   
*Relationsoperatoren:*  $\subseteq$

$$X, Y \subseteq U \vdash \sup_{\mathcal{P}(U), \subseteq} (\{X, Y\}) = X \cup Y$$

|     |   |                                                         |                             |
|-----|---|---------------------------------------------------------|-----------------------------|
| 1   | 1 | $X \subseteq U$                                         | $A$                         |
| 2   | 2 | $Y \subseteq U$                                         | $A$                         |
|     | 3 | $\mathcal{P}(U) \subseteq \mathcal{P}(U)$               | Th. 3.5.3.1                 |
| 1   | 4 | $X \in \mathcal{P}(U)$                                  | Th. 3.16.2.1(1)             |
| 2   | 5 | $Y \in \mathcal{P}(U)$                                  | Th. 3.16.2.1(2)             |
| 1,2 | 6 | $X \cup Y \subseteq U$                                  | Th. 3.13.3.55(1, 2)         |
| 1,2 | 7 | $X \cup Y \in \mathcal{P}(U)$                           | Th. 3.16.2.1(6)             |
|     | 8 | $\text{PartOrd}(\mathcal{P}(U), \subseteq)$             | Th. 14.2.3.1(3)             |
| 1,2 | 9 | $\sup_{\mathcal{P}(U), \subseteq}(\{X, Y\}) = X \cup Y$ | Th. 14.2.3.3(8, 3, 4, 5, 7) |